

INTRODUCTION TO COMPUTER SYSTEM, NETWORKS AND SECURITY

Assignment 1 part 2 - Numbers, Encodings and Boolean logic

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TASK	BRIEF DESCRIPTION
Boolean Algebra Expression	>Write the boolean function as boolean algebra in the form of SOP and POS. >Explain how to get a term from a truth table row.
Logic circuit in Logisim	>Using the boolean function in SOP form, create a single Logisim circuit(using only AND,OR, NOT gates). >Test the circuit for at least 3 test cases.
Optimised circuit	>Create a minimal circuit for the optimised boolean function in SOP form (using only AND,OR, NOT gates). >The optimised boolean function is obtained using karnaugh maps. >Document the karnaugh map process. >Test the minimal circuit for at least 3 test cases.

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BOOLEAN ALGEBRA EXPRESSION

Sum of product:

FOR Z1 OUTPUT:

X1	X2	X3	X4	Z1	PRODUCT
0	0	0	0	0	
0	0	0	1	1	$\overline{X1} \overline{X2} \overline{X3} X4$
0	0	1	0	0	
0	0	1	1	1	$\overline{X1} \overline{X2} X3 X4$
0	1	0	0	1	$\overline{X1} X2 \overline{X3} \overline{X4}$
0	1	0	1	1	$\overline{X1} X2 \overline{X3} X4$
0	1	1	0	1	$\overline{X1} X2 X3 \overline{X4}$
0	1	1	1	0	
1	0	0	0	1	$X1 \overline{X2} \overline{X3} \overline{X4}$
1	0	0	1	1	$X1 \overline{X2} \overline{X3} X4$
1	0	1	0	0	
1	0	1	1	0	
1	1	0	0	1	$X1 X2 \overline{X3} \overline{X4}$
1	1	0	1	0	
1	1	1	0	1	$X1 X2 X3 \overline{X4}$
1	1	1	1	1	$X1 X2 X3 X4$

Therefore the sum of product is:

$$Z1 = \overline{X1} \overline{X2} \overline{X3} X4 + \overline{X1} \overline{X2} X3 X4 + \overline{X1} X2 \overline{X3} \overline{X4} + \overline{X1} X2 \overline{X3} X4 + \overline{X1} X2 X3 \overline{X4} \\ + X1 \overline{X2} \overline{X3} \overline{X4} + X1 \overline{X2} \overline{X3} X4 + X1 X2 \overline{X3} \overline{X4} + X1 X2 X3 \overline{X4} + X1 X2 X3 X4$$

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FOR Z2 OUTPUT:

X1	X2	X3	X4	Z1	PRODUCT
0	0	0	0	1	$\overline{X1} \overline{X2} \overline{X3} \overline{X4}$
0	0	0	1	1	$\overline{X1} \overline{X2} \overline{X3} X4$
0	0	1	0	0	
0	0	1	1	0	
0	1	0	0	0	
0	1	0	1	1	$\overline{X1} X2 \overline{X3} X4$
0	1	1	0	0	
0	1	1	1	1	$\overline{X1} X2 X3 X4$
1	0	0	0	1	$X1 \overline{X2} \overline{X3} \overline{X4}$
1	0	0	1	1	$X1 \overline{X2} \overline{X3} X4$
1	0	1	0	1	$X1 \overline{X2} X3 \overline{X4}$
1	0	1	1	0	
1	1	0	0	1	$X1 X2 \overline{X3} \overline{X4}$
1	1	0	1	1	$X1 X2 \overline{X3} X4$
1	1	1	0	0	
1	1	1	1	1	$X1 X2 X3 X4$

Therefore the sum of product is:

$$Z2 = \overline{X1} \overline{X2} \overline{X3} \overline{X4} + \overline{X1} \overline{X2} \overline{X3} X4 + \overline{X1} X2 \overline{X3} X4 + \overline{X1} X2 X3 X4 + X1 \overline{X2} \overline{X3} \overline{X4} + X1 \overline{X2} \overline{X3} X4 + X1 \overline{X2} X3 \overline{X4} + X1 X2 \overline{X3} \overline{X4} + X1 X2 \overline{X3} X4 + X1 X2 X3 \overline{X4} + X1 X2 X3 X4$$

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Methodology(SOP):

I have broken down the original truth table into two truth tables, one for each of the outputs. Then, from here, for any output that is one(1), the expression is written down. This is known as the product.

NOTE: For SOP;

if $A=1$, it is written as simply A but, if $A=0$ it is written as $\overline{A}$.

For example, let's focus on the table for output Z1;

Row 3 results in a one(1), therefore the product is written down: $\overline{X1} \overline{X2} \overline{X3} X4$.

Now, to obtain the Boolean function for the outputs, all the products are added together, which is quite conveniently called the Sum of Product(SOP).

For example, let's focus on the table for output Z2;

All the products are added together to obtain the SOP:

$$\begin{aligned} &\overline{X1} \overline{X2} \overline{X3} \overline{X4} + \overline{X1} \overline{X2} \overline{X3} X4 + \overline{X1} X2 \overline{X3} X4 + \overline{X1} X2 X3 X4 + X1 \overline{X2} \\ &\overline{X3} \overline{X4} + X1 \overline{X2} \overline{X3} X4 + X1 \overline{X2} X3 \overline{X4} + X1 X2 \overline{X3} \overline{X4} + X1 X2 \overline{X3} X4 + \\ &X1 X2 X3 X4 \end{aligned}$$

Finally the boolean function can be written down:

$$\begin{aligned} Z2 = &\overline{X1} \overline{X2} \overline{X3} \overline{X4} + \overline{X1} \overline{X2} \overline{X3} X4 + \overline{X1} X2 \overline{X3} X4 + \overline{X1} X2 X3 X4 + \\ &X1 \overline{X2} \overline{X3} \overline{X4} + X1 \overline{X2} \overline{X3} X4 + X1 \overline{X2} X3 \overline{X4} + X1 X2 \overline{X3} \overline{X4} + X1 X2 \\ &\overline{X3} X4 + X1 X2 X3 X4 \end{aligned}$$

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Product of Sum:

FOR Z1 OUTPUT:

X1	X2	X3	X4	Z1	SUM
0	0	0	0	0	$X1 + X2 + X3 + X4$
0	0	0	1	1	
0	0	1	0	0	$X1 + X2 + \overline{X3} + X4$
0	0	1	1	1	
0	1	0	0	1	
0	1	0	1	1	
0	1	1	0	1	
0	1	1	1	0	$X1 + \overline{X2} + \overline{X3} + \overline{X4}$
1	0	0	0	1	
1	0	0	1	1	
1	0	1	0	0	$\overline{X1} + X2 + \overline{X3} + X4$
1	0	1	1	0	$\overline{X1} + X2 + \overline{X3} + \overline{X4}$
1	1	0	0	1	
1	1	0	1	0	$\overline{X1} + \overline{X2} + X3 + \overline{X4}$
1	1	1	0	1	
1	1	1	1	1	

Therefore the product of sum is:

$$Z1 = (X1 + X2 + X3 + X4)(X1 + X2 + \overline{X3} + X4)(X1 + \overline{X2} + \overline{X3} + \overline{X4})(\overline{X1} + X2 + \overline{X3} + X4)(\overline{X1} + X2 + \overline{X3} + \overline{X4})(\overline{X1} + \overline{X2} + X3 + \overline{X4})$$

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FOR Z2 OUTPUT:

X1	X2	X3	X4	Z1	SUM
0	0	0	0	1	
0	0	0	1	1	
0	0	1	0	0	$X1 + X2 + \overline{X3} + X4$
0	0	1	1	0	$X1 + X2 + \overline{X3} + \overline{X4}$
0	1	0	0	0	$X1 + \overline{X2} + X3 + X4$
0	1	0	1	1	
0	1	1	0	0	$X1 + \overline{X2} + \overline{X3} + X4$
0	1	1	1	1	
1	0	0	0	1	
1	0	0	1	1	
1	0	1	0	1	
1	0	1	1	0	$\overline{X1} + X2 + \overline{X3} + \overline{X4}$
1	1	0	0	1	
1	1	0	1	1	
1	1	1	0	0	$\overline{X1} + \overline{X2} + \overline{X3} + X4$
1	1	1	1	1	

Therefore the sum of product is:

$$Z2 = (X1 + X2 + \overline{X3} + X4)(X1 + X2 + \overline{X3} + \overline{X4})(X1 + \overline{X2} + X3 + X4)(X1 + \overline{X2} + \overline{X3} + X4)(\overline{X1} + X2 + \overline{X3} + \overline{X4})(\overline{X1} + \overline{X2} + \overline{X3} + X4)$$

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Methodology(POS):

For any output that is zero(0), the expression is written down. This is known as the sum.

NOTE: For POS;

If A=1, it is written as simply $\bar{A}$ but, if A=0 it is written as A.

For example, let's focus on the table for output Z1;

Row 1 results in a zero(0), therefore the sum is written down:

$$X1 + X2 + X3 + X4.$$

Now, to obtain the Boolean function for the outputs, all the sums are multiplied together, which is quite conveniently called the Product of Sum(POS).

For example, let's focus on the table for output Z2;

All the products are added together to obtain the POS:

$$(X1 + X2 + \bar{X3} + X4)(X1 + X2 + \bar{X3} + \bar{X4})(X1 + \bar{X2} + X3 + X4)(X1 + \bar{X2} + \bar{X3} + X4)(\bar{X1} + X2 + \bar{X3} + \bar{X4})(\bar{X1} + \bar{X2} + \bar{X3} + X4)$$

Finally the boolean function can be written down:

$$Z2 = (X1 + X2 + \bar{X3} + X4)(X1 + X2 + \bar{X3} + \bar{X4})(X1 + \bar{X2} + X3 + X4)(X1 + \bar{X2} + \bar{X3} + X4)(\bar{X1} + X2 + \bar{X3} + \bar{X4})(\bar{X1} + \bar{X2} + \bar{X3} + X4)$$

A brief observation: SOP and POS are the opposite of each other.

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LOGICAL CIRCUIT IN LOGISIM

Brief explanation of the circuit construction:

From the boolean function obtain, the circuit is obstructed as follows;

- >Each term in the “product” is connected to an AND gate,
- >Since both outputs(Z1 & Z2) have 10 products, there are 10 AND gates for both.
- >Then all the outputs of the 10 respective AND gates of each circuit are connected to an OR gate which outputs the result.

Testing:

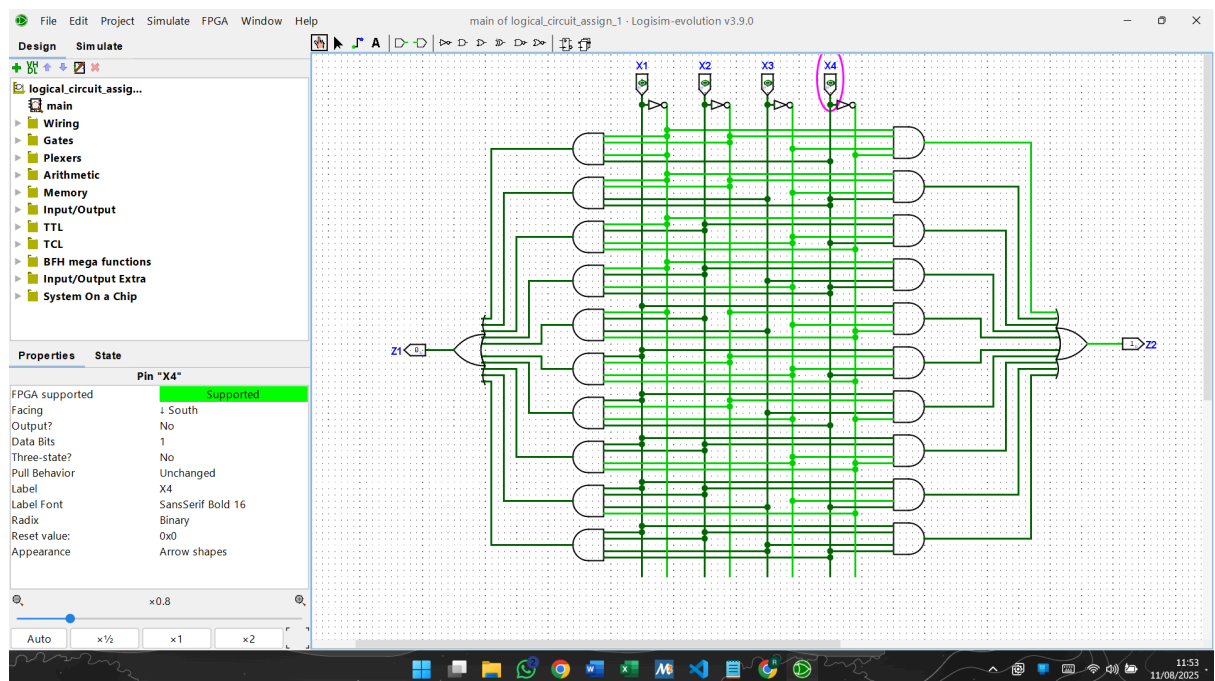
1st test(common bit testing):

For this test, I checked for the output values for when all the input bits are the same.

I.e. Input value: 0000

Expected output for Z1: 0

Expected output for Z2: 1



Actual output for Z1: 0

Actual output for Z2: 1

Since the expected outputs and the actual outputs match, the circuits are correct.

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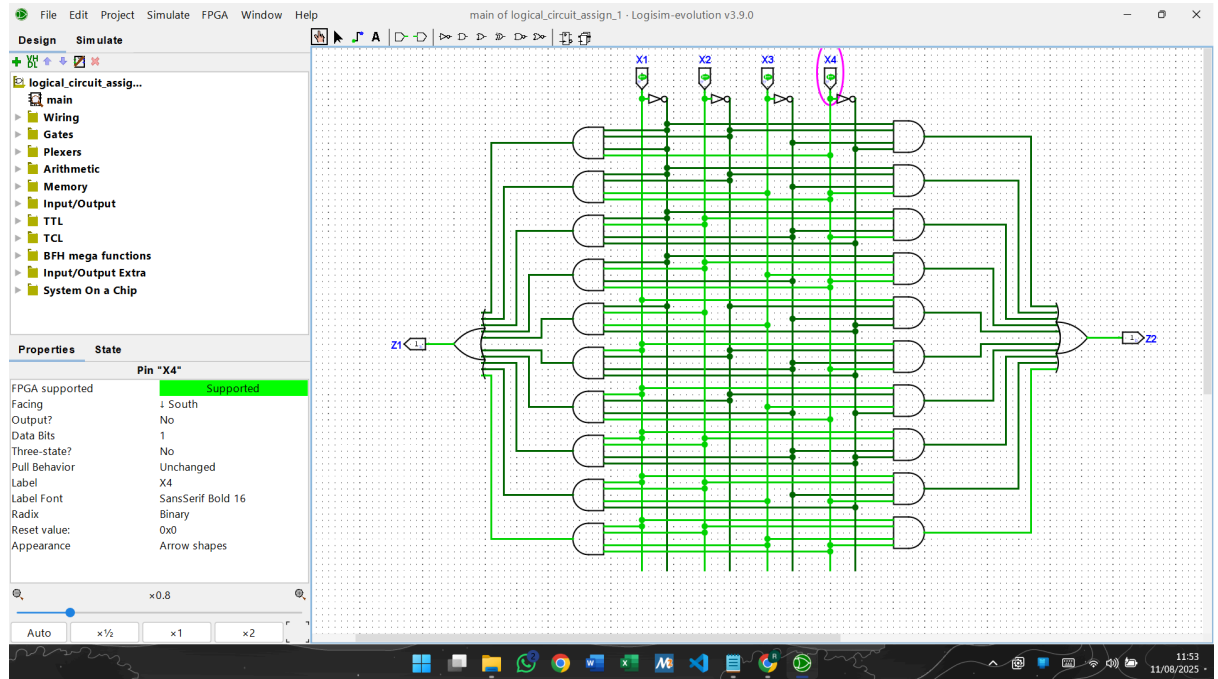
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I.e. Input value: 1111

Expected output for Z1: 1

Expected output for Z2: 1



Actual output for Z1: 1

Actual output for Z2: 1

Since the expected outputs and the actual outputs match, the circuits are correct.

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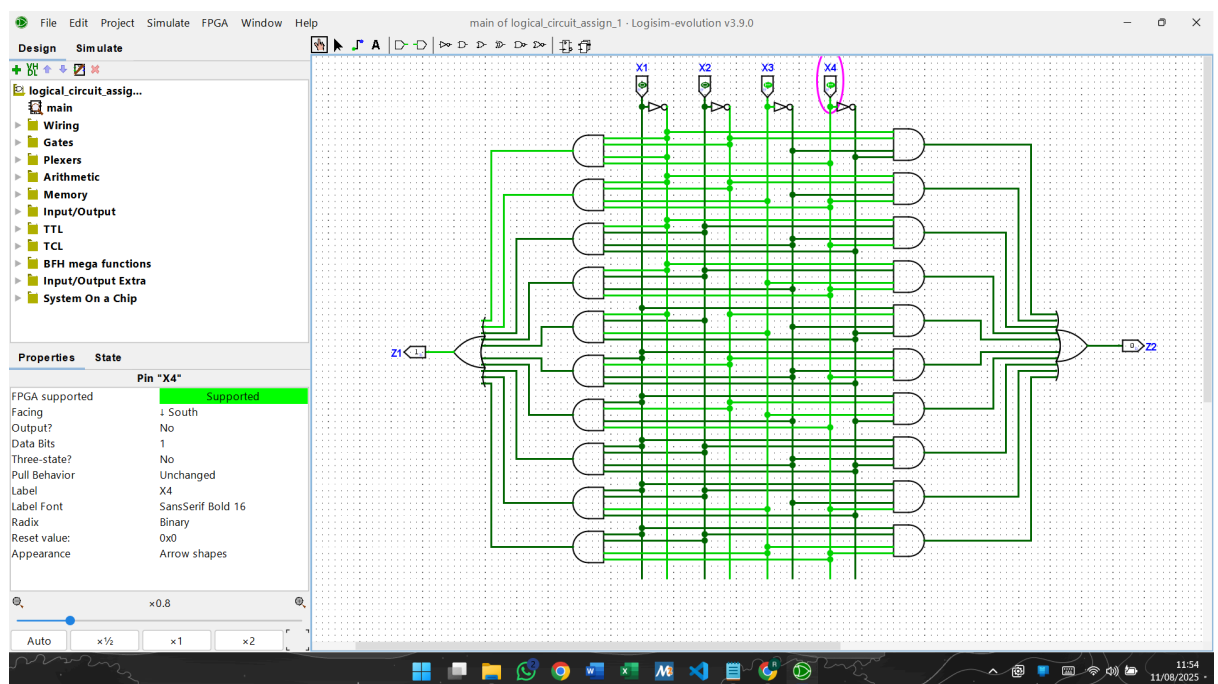
2nd test(Random testing):

For this test, random inputs that are not 0000 or 1111 are chosen and the expected outputs and actual outputs are compared.

Random input: 0011

Expected output for Z1: 1

Expected output for Z2: 0



Actual output for Z1: 1

Actual output for Z2: 0

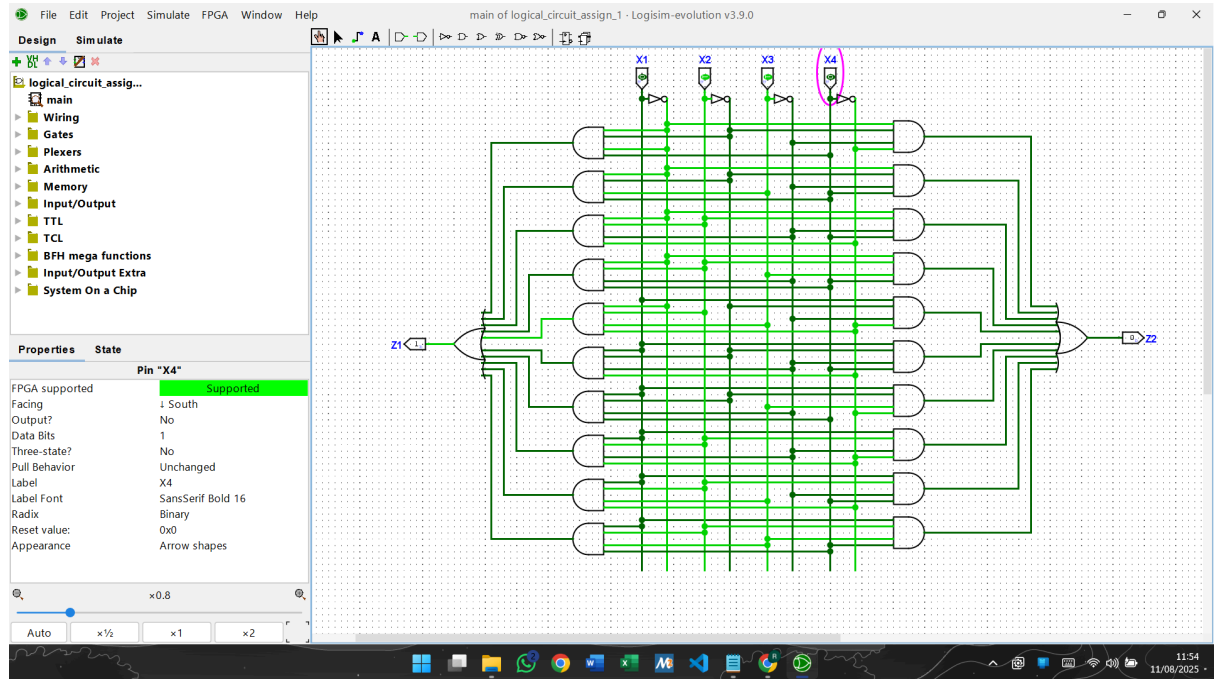
Since the expected outputs and the actual outputs match, the circuits are correct.

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Random input: 0110
Expected output for Z1: 1
Expected output for Z2: 0



Actual output for Z1: 1
Actual output for Z2: 0

Since the expected outputs and the actual outputs match, the circuits are correct.

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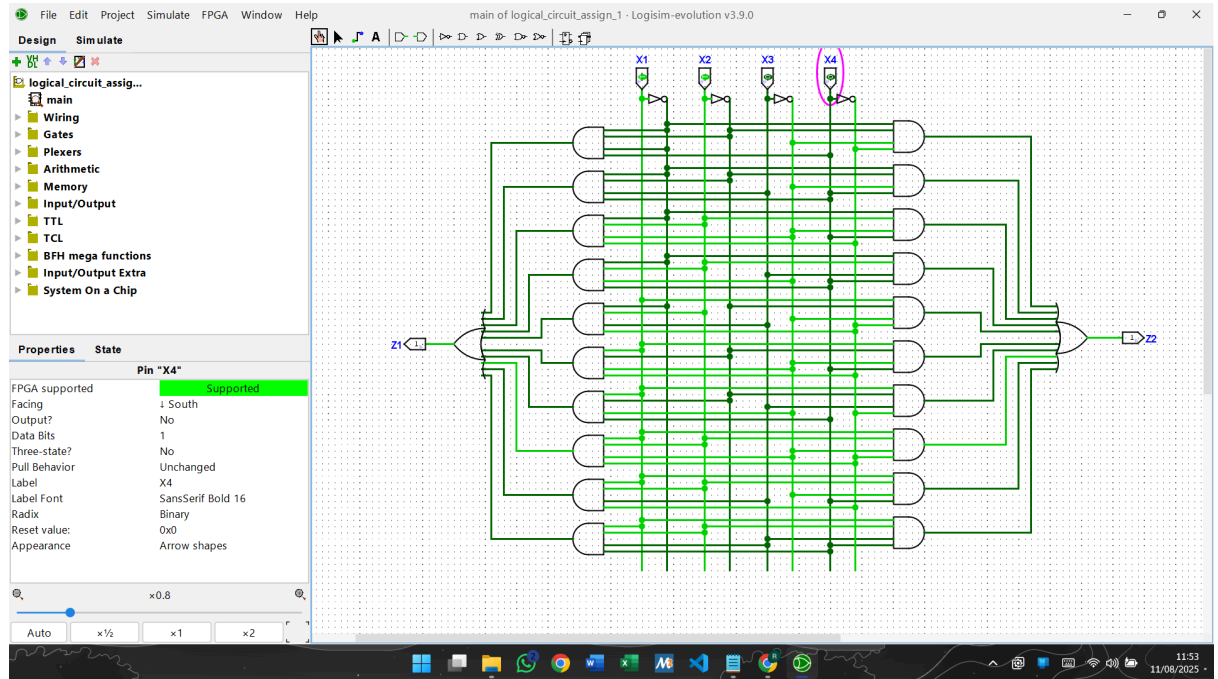
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Random input: 1100

Expected output for Z1: 1

Expected output for Z2: 1



Actual output for Z1: 1

Actual output for Z2: 1

Since the expected outputs and the actual outputs match, the circuits are correct.

Therefore from the above tests, I can conclude that the circuits are indeed correct.

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OPTIMISED CIRCUIT

For Z1 circuit:

Karnaugh Map:

$X_3X_4 \backslash X_1X_2$	00	01	11	10
00	0	1	1	0
01	1	1	0	1
11	1	0	1	1
10	1	1	0	0

Therefore, the optimised function is:

$$Z1 = \overline{X_2} \overline{X_4} + \overline{X_1} \overline{X_2} X_4 + \overline{X_1} \overline{X_2} \overline{X_3} + \overline{X_1} X_2 \overline{X_3} + X_1 X_2 X_3$$

For Z2 circuit:

Karnaugh Map:

$X_3X_4 \backslash X_1X_2$	00	01	11	10
00	1	1	0	0
01	0	1	1	0
11	1	1	1	0
10	1	1	0	1

Therefore, the optimised function is:

$$Z2 = \overline{X_2} \overline{X_3} + \overline{X_1} \overline{X_3} + X_2 X_4 + \overline{X_1} \overline{X_2} \overline{X_4}$$

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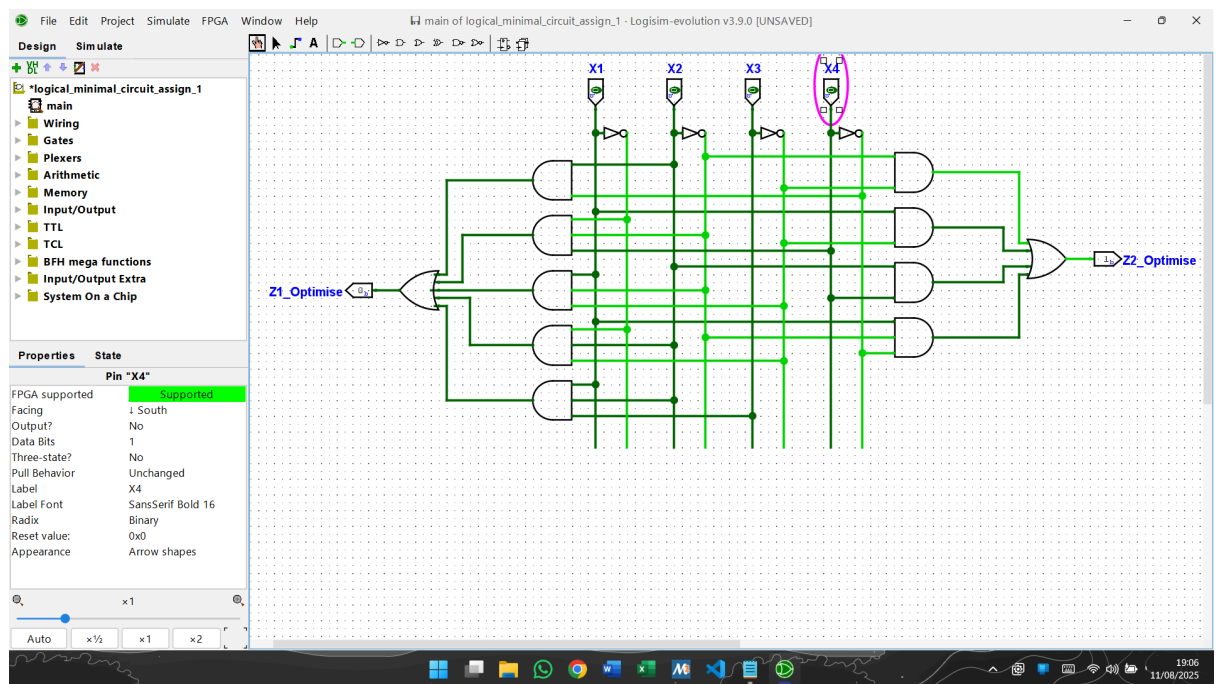
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OPTIMISED CIRCUIT IN LOGISIM

THE CIRCUIT:

The following is a picture of the optimised circuits;



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Testing:

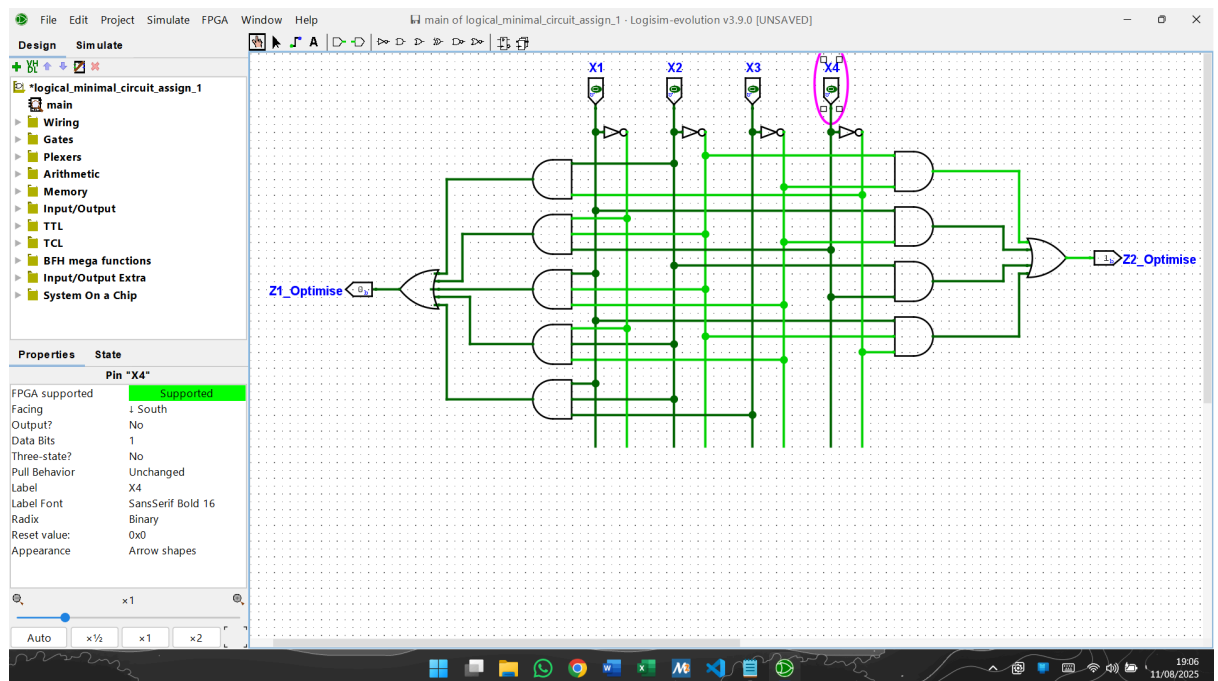
1st test(common bit testing):

For this test, I checked for the output values for when all the input bits are the same.

I.e. Input value: 0000

Expected output for Z1: 0

Expected output for Z2: 1



Actual output for Z1: 0

Actual output for Z2: 1

Since the expected outputs and the actual outputs match, the circuits are correct.

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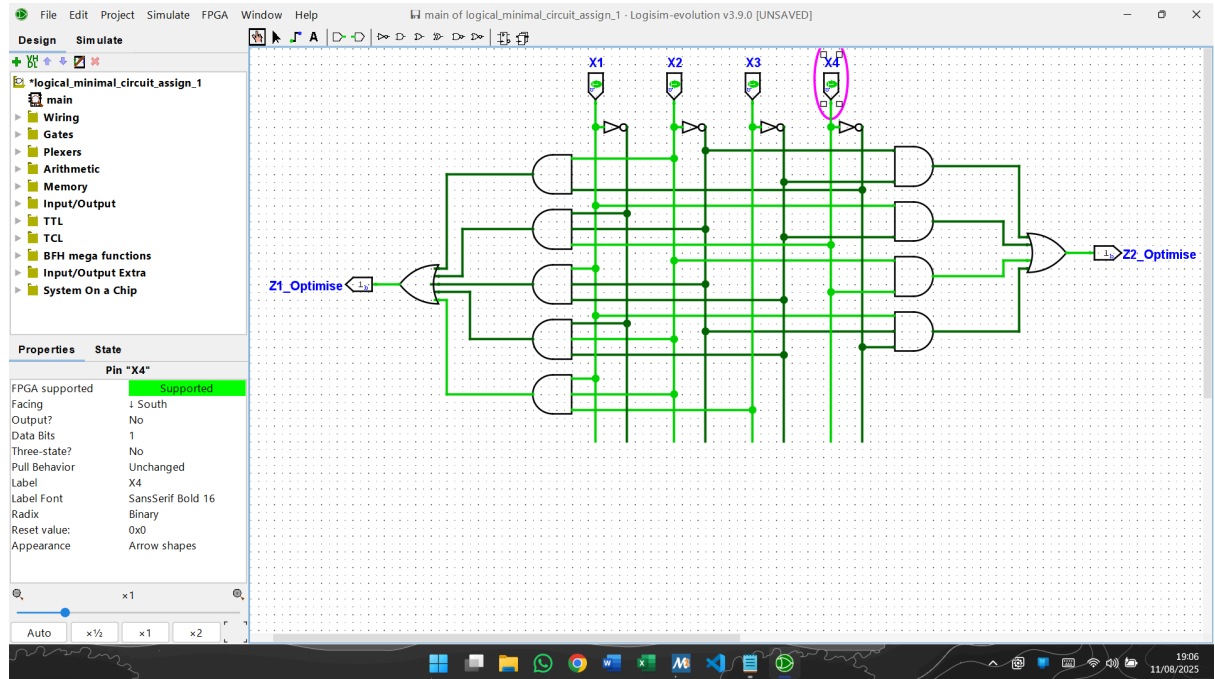
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I.e. Input value: 1111

Expected output for Z1: 1

Expected output for Z2: 1



Actual output for Z1: 1

Actual output for Z2: 1

Since the expected outputs and the actual outputs match, the circuits are correct.

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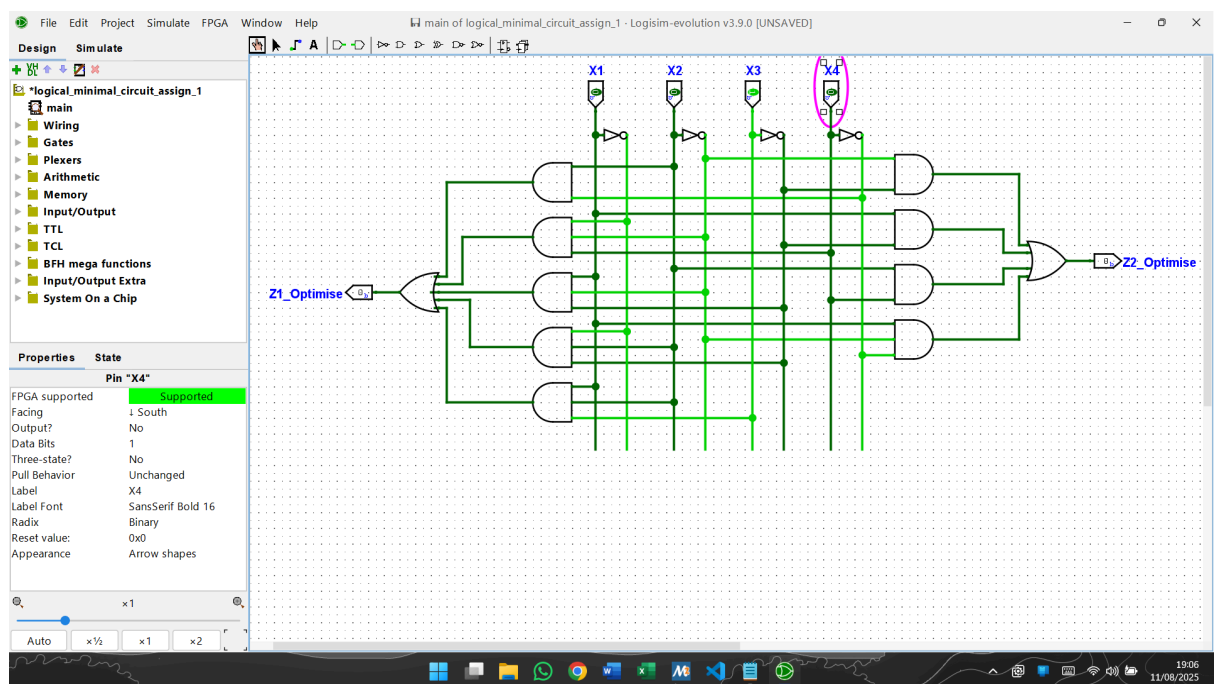
2nd test(Random testing):

For this test, random inputs that are not 0000 or 1111 are chosen and the expected outputs and actual output are compared.

Random input: 0010

Expected output for Z1: 0

Expected output for Z2: 0



Actual output for Z1: 0

Actual output for Z2: 0

Since the expected outputs and the actual outputs match, the circuits are correct.

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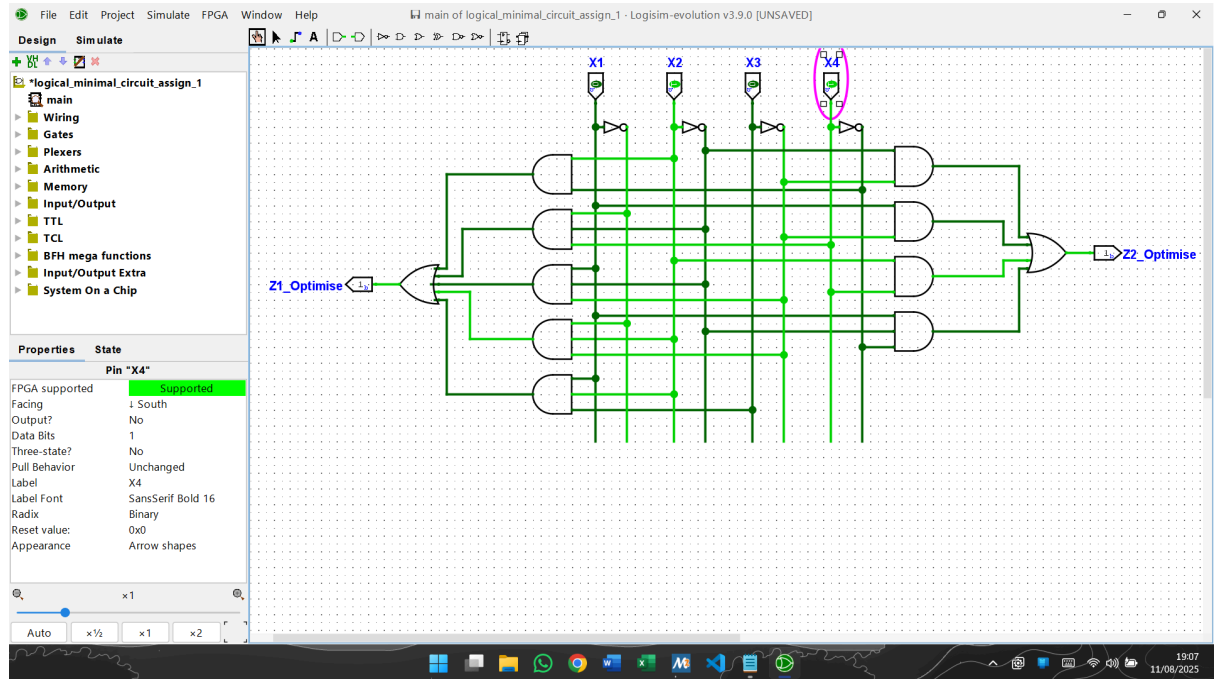
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Random input: 0101

Expected output for Z1: 1

Expected output for Z2: 1



Actual output for Z1: 1

Actual output for Z2: 1

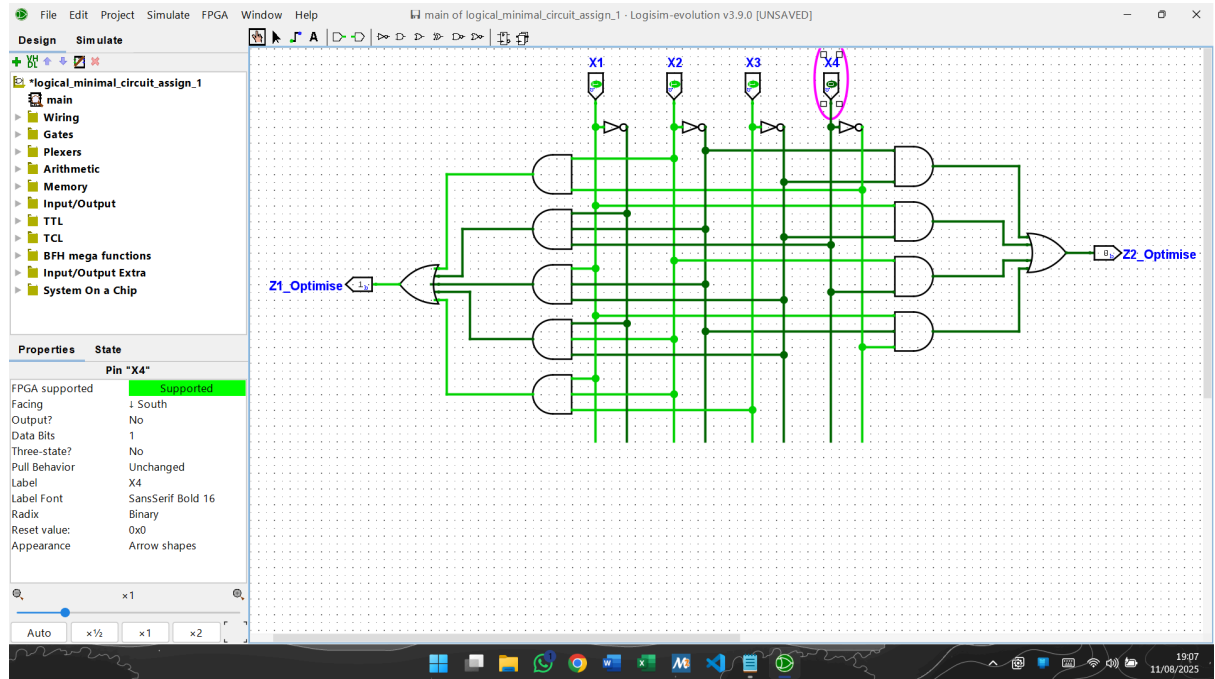
Since the expected outputs and the actual outputs match, the circuits are correct.

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Random input: 1110
Expected output for Z1: 1
Expected output for Z2: 0



Actual output for Z1: 1
Actual output for Z2: 0

Since the expected outputs and the actual outputs match, the circuits are correct.